

Reduction of unsaturated compounds

Learning outcomes

By the end of this section you should be able to deduce the products of the reactions of hydrogen with alkenes and alkynes.

Having a balanced diet is essential for good health. Humans need a diet filled with the main groups of biomolecules (carbohydrates, lipids, and proteins), and essential nutrients, vitamins and minerals. Nowadays, there are a wide variety of oils and fats collected or produced for human consumption. For a few decades in some countries, margarine has been used as a mainstream alternative to butter, as it is low cost and is marketed as a healthier fat and a vegetarian/vegan option or suitable for those with dairy allergies.

Margarine is a solid fat produced industrially from vegetable oils that are liquid at room temperature, however, margarine is solid. What distinguishes margarine from other plant oils and what allows it to be solid at room temperature?



Figure 1. Margarine is a solid substance at room temperature.

Credit: Science Photo Library, Getty Images

Reduction of unsaturated compounds

Recall the structure of unsaturated compounds (see subtopic S3.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43122/>)). Unsaturated compounds are hydrocarbons that possess at least one double or triple carbon-carbon bond in their main chain. Compounds with double bonds are called alkenes and compounds with triple bonds are called alkynes.

To identify an unsaturated compound, an unsaturation test can be performed that consists of adding bromine water and checking for a colour change. This reaction is classified as a halogenation, as the double bond in an alkene (or the triple bond in an alkyne) will react with the bromine, $\text{Br}_2(\text{l})$, and immediately change its colour from brown to colourless. When bromine is mixed with an alkane, there is no colour change.

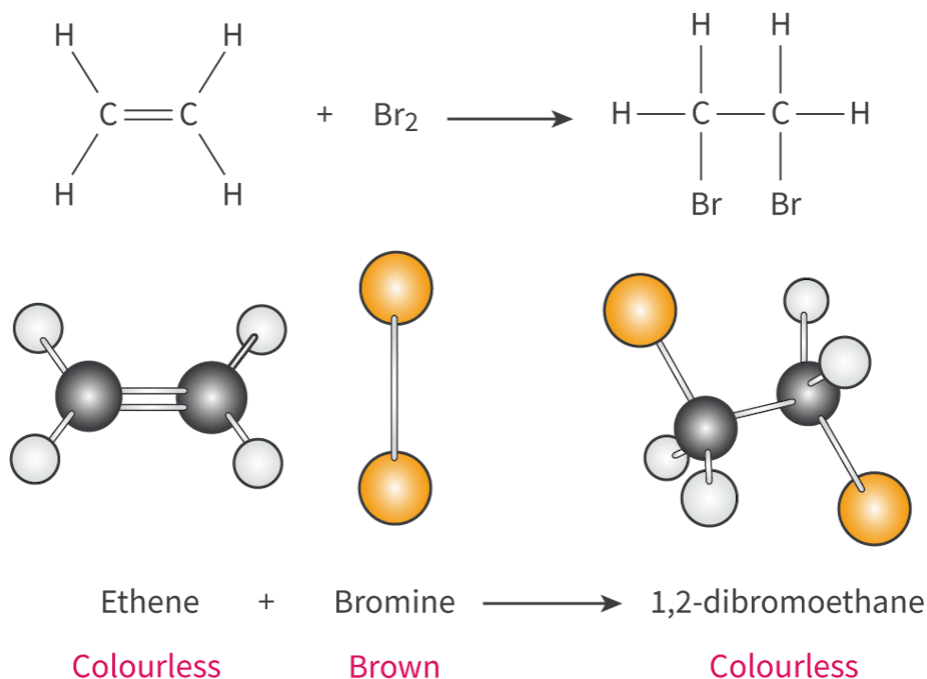


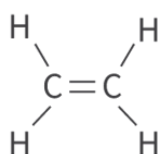
Figure 2. The halogenation of ethene.

 More information for figure 2

The image is a chemical diagram illustrating the halogenation reaction of ethene with bromine to form 1,2-dibromoethane. At the top, there is a balanced chemical equation: $(\text{C}_2\text{H}_4 + \text{Br}_2 \rightarrow \text{C}_2\text{H}_4\text{Br}_2)$. Below the equation, there is a structural representation of ethene on the left, depicted as a double-bonded carbon chain with hydrogen atoms attached, labeled "Colourless" underneath. In the center, there is a model of bromine as a diatomic molecule represented by two orange spheres, labeled "Brown." On the right, there is a newly formed molecule of 1,2-dibromoethane shown with the addition of two bromine atoms to the original ethene structure, labeled "Colourless." These visual representations are connected by an arrow, indicating the chemical reaction.

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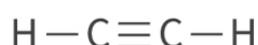
Look at the structure of the alkenes and alkynes in terms of potential oxidation. Which reactions are possible?



 More information

The image is a structural diagram of an ethylene molecule. It consists of two carbon (C) atoms connected by a double bond, represented by two parallel lines between the carbon atoms. Each carbon atom is also bonded to two hydrogen (H) atoms. The hydrogen atoms are positioned in a manner that creates a linear and symmetrical arrangement around the carbon atoms, forming a planar geometry. The bonds between carbon and hydrogen are represented by lines extending outward from each carbon atom towards the hydrogen atoms, creating a structural formula C_2H_4 .

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 More information

This image shows the structural formula of Ethyne (alkyne), also known as acetylene. It consists of two carbon atoms (C) connected by a triple bond, represented by three parallel lines. Each carbon atom is additionally bonded to a hydrogen atom (H) on the outer sides with single bonds. The overall structure is linear, with $H-C \equiv C-H$, indicating a bond between hydrogen and carbon atoms on both ends of the carbon triple bond.

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Ethene
(alkene)

Ethyne
(alkyne)

Figure 3. Alkenes and alkynes.

You might have thought that both alkenes and alkynes can be easily reduced by the addition of hydrogen atoms, when the double or triple bonds are broken. The process of adding hydrogens is called hydrogenation. This reaction decreases the degree of unsaturation of a compound. The degree of unsaturation is a measurement of the number of molecules of hydrogen (H_2) needed to convert the organic compound to the corresponding saturated, non-cyclic molecule.

In the hydrogenation of an alkene, gaseous hydrogen (H_2) is added across the carbon-carbon double bond. The product of the reaction is an alkane, a saturated hydrocarbon. Note that the conditions for this reaction are high pressure and temperature with a nickel (Ni) catalyst.

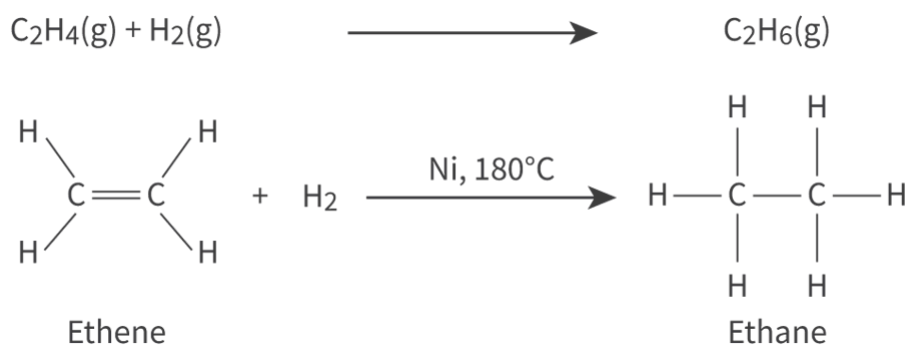


Figure 4. The hydrogenation of ethene.

 More information for figure 4

The image depicts a chemical equation representing the hydrogenation of ethene (C_2H_4) to ethane (C_2H_6). On the left side of the equation, the structure of ethene is shown with two carbon atoms double-bonded to each other and also bonded to two hydrogen atoms each. Gaseous hydrogen (H_2) is added to this structure. The direction of the reaction is indicated by an arrow pointing to the right. Above the arrow, it mentions that a nickel catalyst and a temperature of 180°C are required for this reaction. On the right side, the structure of ethane is displayed, consisting of two carbon atoms single-bonded to each other and each carbon atom bonded to three hydrogen atoms. This transformation results in a saturated hydrocarbon, as indicated by the change from the double bond in ethene to a single bond in ethane.

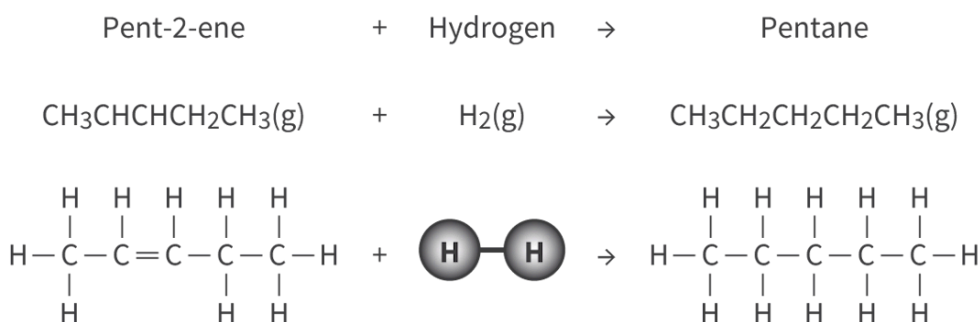
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Unsaturated compounds are bent due to the double and triple bonds in their chains. They are not easily stacked up when close to each other, which means they have reduced London forces when compared to the saturated hydrocarbon. Weaker London forces means a lower melting and boiling point. Vegetable oils, being composed of unsaturated lipids, have lower melting points and are, therefore, liquid at room temperature. In the production of margarine, the unsaturated compounds undergo a reduction reaction and become saturated, which means they will no longer be bent. Becoming straight chained, the compounds have stronger London forces between molecules, which leads to a higher melting point. This allows margarine to be solid at room temperature.

Worked example 1

Draw the overall reaction and name the product of the hydrogenation of pent-2-ene.

In the hydrogenation of pent-2-ene, the double bond is broken and each hydrogen takes one of the unpaired electrons from each of the carbons 2 and carbon 3 atoms, to produce pentane.



The hydrogenation of alkynes involves the addition of hydrogen gas (H_2) to a carbon-carbon triple bond, resulting in the formation of a carbon-carbon double bond or a single bond, depending on the reaction conditions.

For example, in the hydrogenation of ethyne, ethyne is converted into ethene by the addition of hydrogen gas. The reaction is typically carried out in the presence of a metal catalyst, such as palladium, platinum, or nickel.

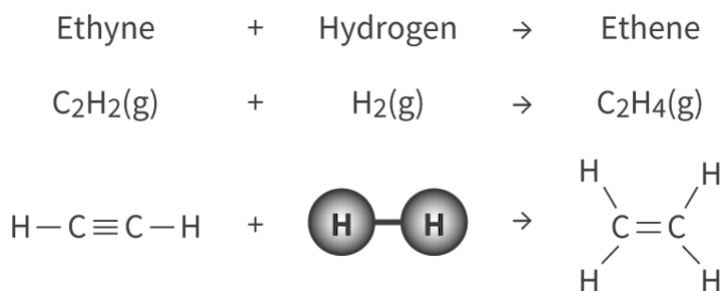


Figure 5. The hydrogenation of ethyne to ethene.

 More information for figure 5

The image is a diagram illustrating the chemical process of the hydrogenation of ethyne into ethene. At the top, it shows the words: "Ethyne + Hydrogen → Ethene." Below it, the chemical formulas are displayed: " $\text{C}_2\text{H}_2(\text{g}) + \text{H}_2(\text{g}) \rightarrow \text{C}_2\text{H}_4(\text{g})$." The image visually represents the structural formulas of the molecules involved. It features a linear ethyne molecule with a triple bond between carbon atoms, represented as " $\text{H}-\text{C}\equiv\text{C}-\text{H}$." This molecule is on the left side of the equation. It then shows a diatomic hydrogen molecule (H_2), represented as two connected spheres, indicating a single bond, located in the middle of the equation. On the right side, the ethene molecule is shown, with a double bond between the carbon atoms: " $\text{C}=\text{C}$," with two hydrogen atoms bonded to each carbon atom, resulting in the formula " $\text{H}_2\text{C}=\text{CH}_2$." This structural transformation visually emphasizes the addition of hydrogen to the ethyne molecule to form ethene, illustrating the change from a triple to a double bond between carbon atoms.

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Alkynes can also be reduced to alkanes. In **Figure 6**, two molecules of hydrogen gas (H_2) are added to one molecule of ethyne (C_2H_2) to form one molecule of ethane (C_2H_6). The reaction is typically carried out in the presence of a metal catalyst, such as palladium or platinum, at high temperature and pressure.

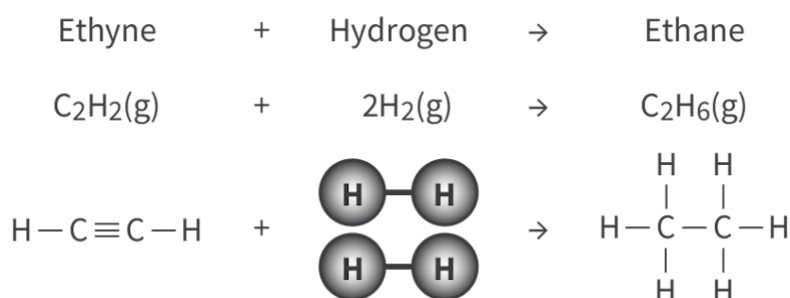


Figure 6. The hydrogenation of ethyne to ethane.

 More information for figure 6

The image is a chemical reaction diagram illustrating the hydrogenation of ethyne into ethane. It shows the chemical equation as follows: Ethyne (C_2H_2) plus two Hydrogen (H_2) molecules yield Ethane (C_2H_6).

On the left, the diagram displays a structural formula for ethyne, written as $\text{H}-\text{C}\equiv\text{C}-\text{H}$, indicating a triple bond between the carbon atoms. Next to it are two separate H_2 molecules represented by two pairs of ball-and-stick models, where each pair of balls represents a hydrogen molecule.

An arrow points to the products on the right side of the diagram, symbolizing the chemical reaction. Here, the ethane molecule is displayed as a structural formula: H-C-C-H, with single hydrogen atoms bonded to each carbon, forming a complete ethane structure with a single bond between the carbon atoms.

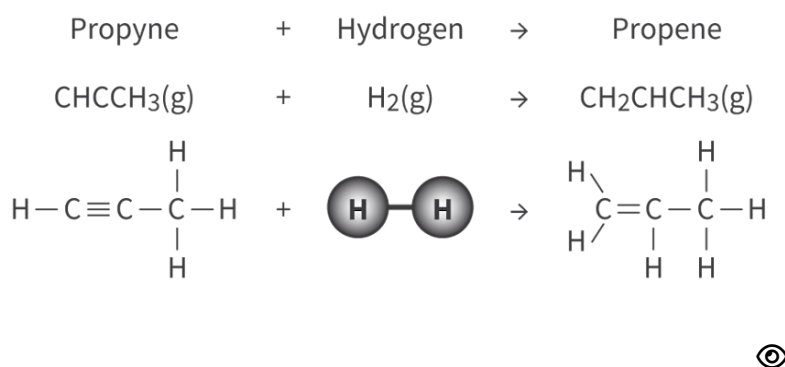
The diagram visually depicts the transformation involving the breaking of the triple bond and the formation of a single bond with the addition of hydrogen atoms, showing the transition from an alkyne to an alkane.

[Generated by AI]

Worked example 2

Draw the overall reaction and name the product of the hydrogenation of propyne (note: the final product has a C=C bond).

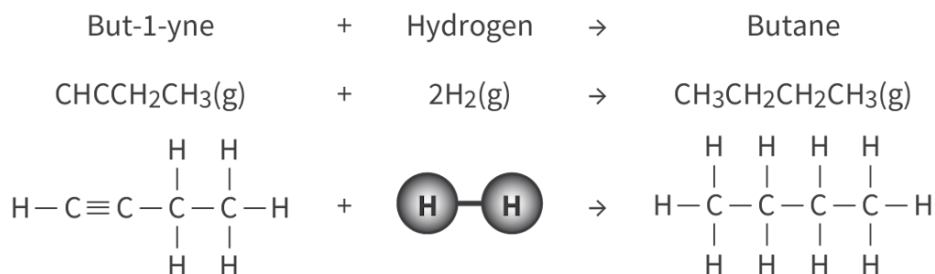
In the hydrogenation of propyne, the triple bond is broken and each hydrogen takes one of the unpaired electrons of carbons 1 and 2, producing propene. A double bond is still left.



Worked example 3

Draw the overall reaction and name the product of the hydrogenation of but-1-yne (note: the final product has only C-C bonds).

In the hydrogenation of but-1-yne, the triple bond is broken and each hydrogen takes one of the unpaired electrons of the carbon 1 and carbon 2 atoms, to produce butane.



Making connections

Why are some reactions of alkenes classified as reduction reactions while others are classified as electrophilic addition reactions? (see [subtopic R3.4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43118/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43118/>))

The distinction between a reduction and an electrophilic addition reaction of alkenes depends on the molecule reacting with the alkene and the reaction mechanism.

In a reduction reaction, like the ones just studied, the alkene reacts with a molecule of hydrogen that can donate hydrogens (reduction), resulting in the decrease of the oxidation state of the carbons involved with the double bond, and the production of an alkane.

In an electrophilic addition (see [section R3.4.4—5](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrophilic-addition-reaction-id-46329/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/electrophilic-addition-reaction-id-46329/>)), an electrophile forms a bond to the nucleophile (in this case the C=C) by accepting both bonding electrons from the carbon in the C=C. Common electrophiles include hydrogen halides (H—X), halogens (X₂), and polar molecules such as water (H₂O) and alcohols (—OH). Electrophilic addition reactions result in the formation of a single bond between the two carbon atoms and a new functional group on the molecule.

Activity

- **IB learner profile attribute:** Thinkers
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual/pair activity

Reduction of unsaturated compounds by hydrogenation

Materials

Paper, pencil, molecular model kit (optional)

Your task

1. Review the definition and examples of unsaturated compounds and degree of unsaturation. Recall that unsaturated compounds contain double or triple bonds between carbon atoms, and that the degree of unsaturation is the measurement of the number of hydrogen molecules (H_2) needed to convert the organic compound to the corresponding saturated, non-cyclic molecule.
2. On a piece of paper, draw the structures of three unsaturated compounds with different degrees of unsaturation: ethene (C_2H_4), propyne (C_3H_4) and cyclohexene (C_6H_{10}).
3. For each compound, write the balanced equation for its hydrogenation reaction. Draw the structure of the product.
4. Compare the degrees of unsaturation of the reactants and products. How does hydrogenation affect the degree of unsaturation? Explain your answer using chemical concepts and terminology.
5. Choose one of the compounds and research its uses and properties. How would hydrogenation change its uses and properties?
6. Write a paragraph summarising your findings and citing your sources.
7. Share your paragraph with a classmate and discuss your findings.